Managed DevOps Pools:
Elastische Azure DevOps Agents - ganz ohne Ops

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empower your software solutions



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Our Products:

Multi-Tenant OpenID Connect Identity Provider

ProAuth

www.proauth.net

Enterprise Application Framework for .NET

ReaFx

www.reafx.net

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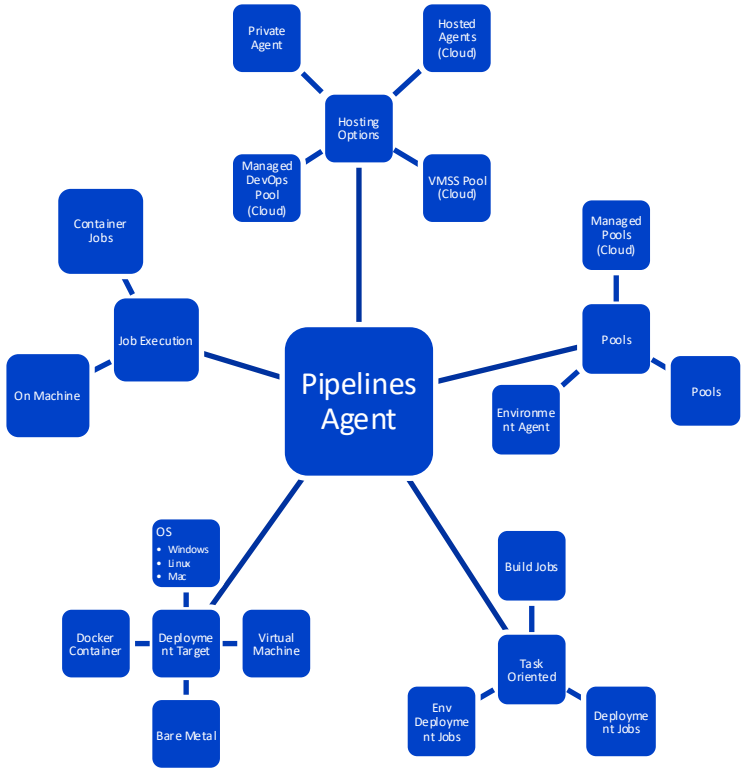
https://www.4tecture.ch/events/dwx26-mdp

ONE AGENT TO RULE THEM ALL...

The Agent

- Multi-purpose task executer
- Executes any task
- Orchestration by Azure Pipelines

Azure Pipelines Agents



Considerations / Challenges

- Tool Dependencies
- Build old and new states of the repo
- Network access / restrictions
- Scaling
- Caching

HOSTED AGENT VS. PRIVATE AGENT

Agent Hosting Options

Hosted Agent	Virtual Machine Scale Set Pool	Managed DevOps Pool	Private Agent
• Cloud-Only • Fixed image for Windows, Linux, Mac • Monthly fee per parallel agent instance • Job duration limits	• Cloud-Only • Run agents in your VMSS resource • Scaled by Azure DevOps Services • You provide VM images • Configurable VM resources	• Cloud-Only • Hosted on your behalf • Quick-starter images and custom images • Stateful agents • Private networking • Configurable VM resources	• Runs everywhere • Self-installed • Self-scaled

Demo



Managed DevOps Pools

Managed DevOps Pools

- Microsoft VM images and custom VM images
- Any VM type and size
- vNet Integration
- Managed Identity Support
- Billing through Microsoft Azure Subscription
- Sophisticated Scaler

Demands – Work Folder

```
pool:
  name: fabrikam-managed-pool # Name of pool
  demands:
  - WorkFolder -equals /mnt/storage/sdc/custom-work-folder
```

```
pool:
  name: fabrikam-managed-pool # Name of pool
  demands:
  - WorkFolder -equals f:\custom-work-folder
```

Demands – Priority

```
pool:
  name: fabrikam-dev-pool # Name of pool
  demands:
  - Priority -equals Low
```

Demands – Image Override

```yaml
pool:
  name: fabrikam-dev-pool # Name of pool
  demands:
  - ImageOverride -equals ubuntu-24.04-gen2
```

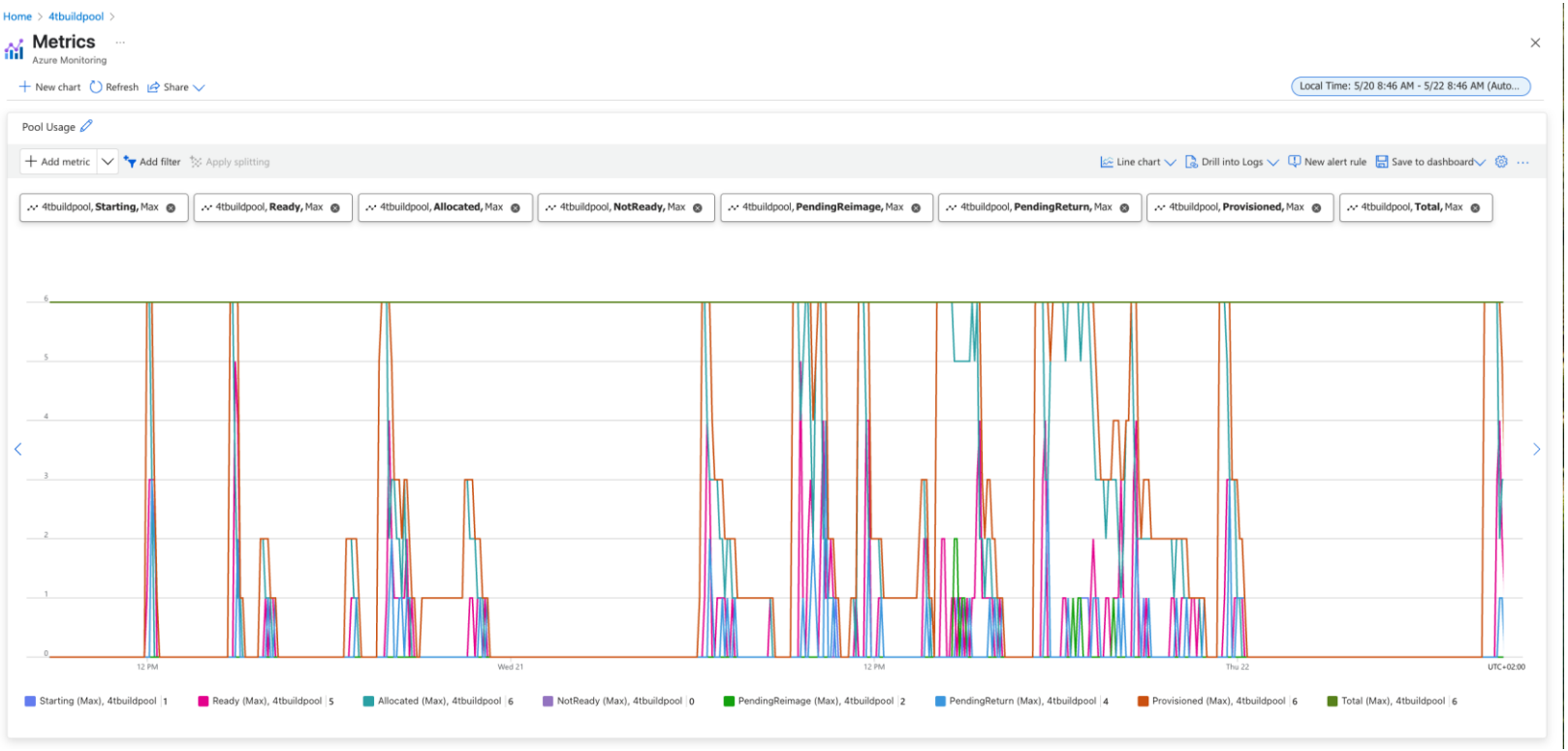
Demands – Image Version Override

```yaml
pool:
  name: fabrikam-dev-pool # Name of pool
  demands:
  - ImageVersionOverride -equals 20250427.1.0
```

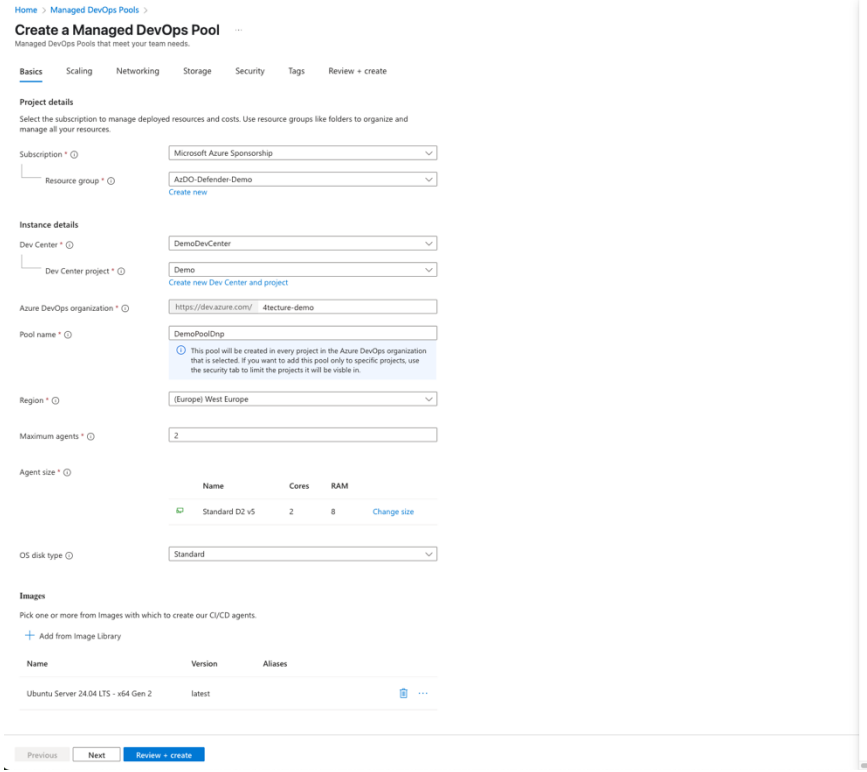
Demands – Custom Capabilities

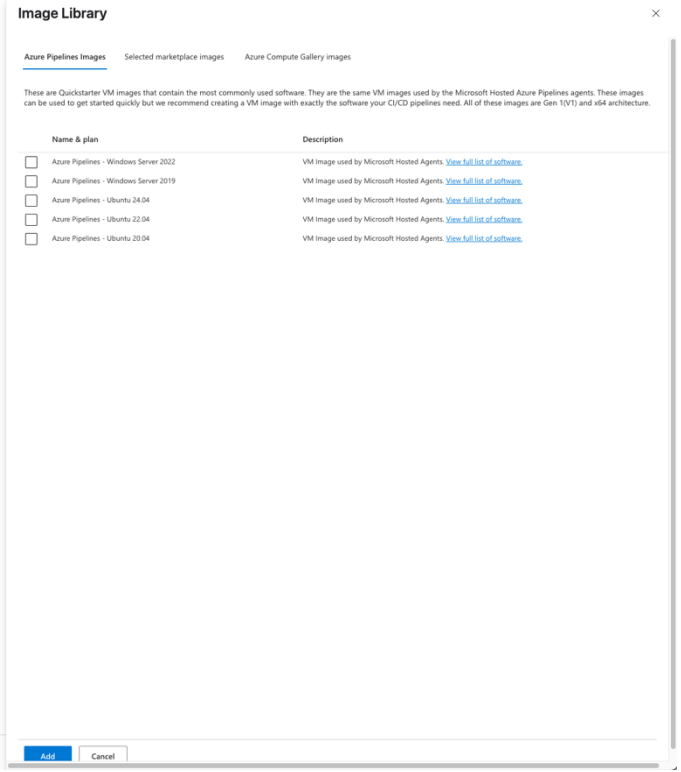
```yaml
pool:
  name: fabrikam-dev-pool # Name of pool
  demands:
  - ImageOverride -equals windows-2022
  - CustomCapabilities -equals MyCustomValue
```

Metrics

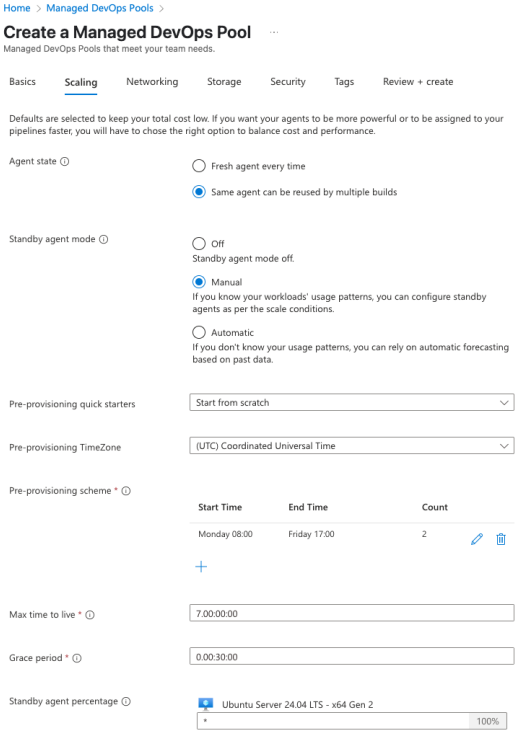

Home > 4tbuildpool >
Metrics
Azure Monitoring
New chart
Refresh
Share
Local Time: 5/20 8:46 AM - 5/22 8:46 AM (Auto...
Pool Usage
Add metric
Add filter
Apply splitting
Line chart
Drill into Logs
New alert rule
Save to dashboard
4tbuildpool, Starting, Max
4tbuildpool, Ready, Max
4tbuildpool, Allocated, Max
4tbuildpool, NotReady, Max
4tbuildpool, PendingReimage, Max
4tbuildpool, PendingReturn, Max
4tbuildpool, Provisioned, Max
4tbuildpool, Total, Max
6
5
4
3
2
1
0
12 PM
Wed 21
12 PM
Thu 22
UTC+02:00
Starting (Max), 4tbuildpool 1
Ready (Max), 4tbuildpool 5
Allocated (Max), 4tbuildpool 6
NotReady (Max), 4tbuildpool 0
PendingReimage (Max), 4tbuildpool 2
PendingReturn (Max), 4tbuildpool 4
Provisioned (Max), 4tbuildpool 6
Total (Max), 4tbuildpool 6

Configuration

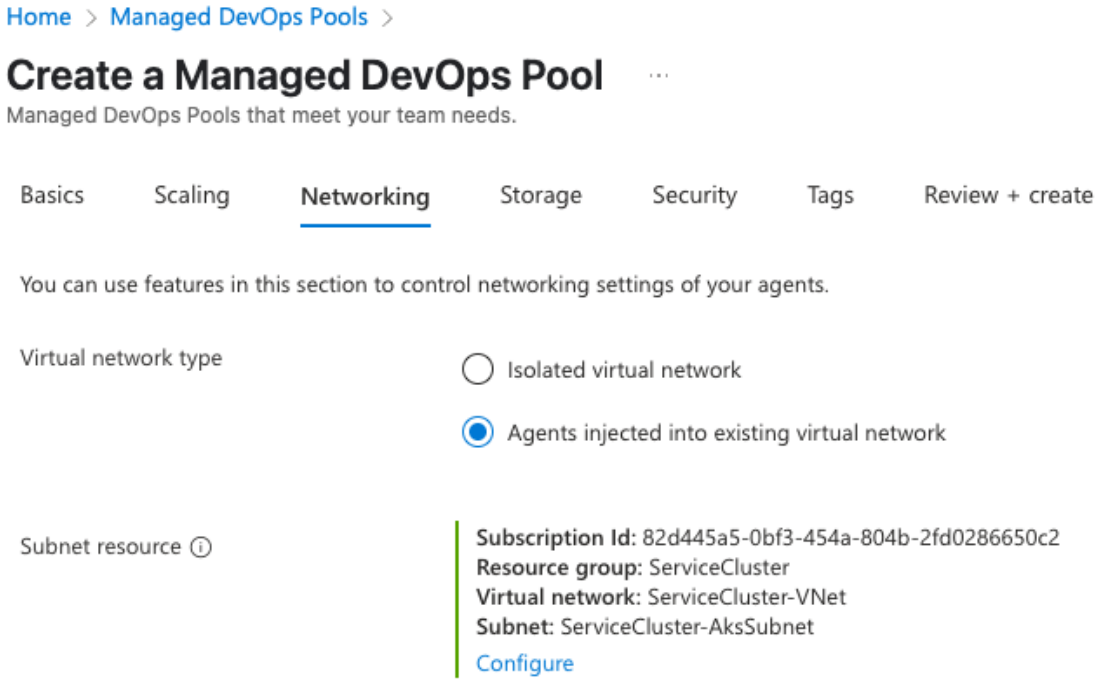




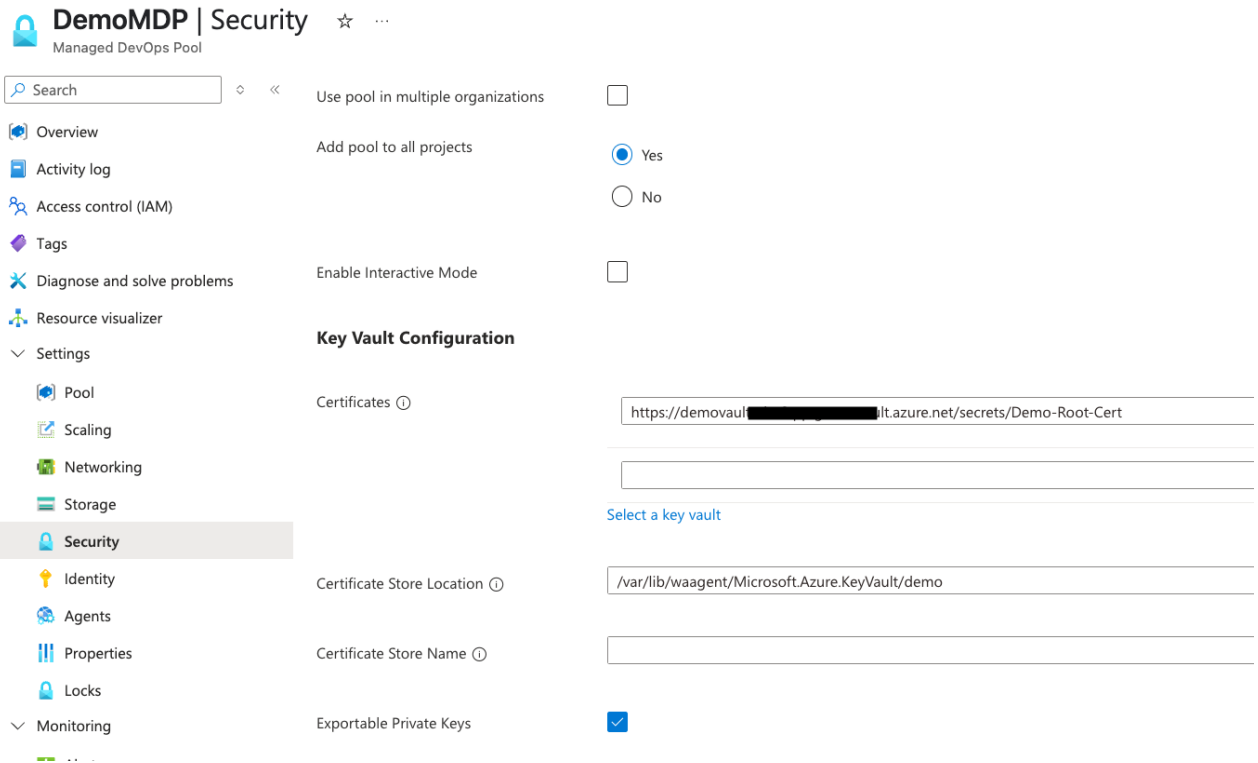
Scaling


Home > Managed DevOps Pools >
Create a Managed DevOps Pool
Managed DevOps Pools that meet your team needs.
Basics
Scaling
Networking
Storage
Security
Tags
Review + create
Defaults are selected to keep your total cost low. If you want your agents to be more powerful or to be assigned to your pipelines faster, you will have to chose the right option to balance cost and performance.
Agent state
Fresh agent every time
Same agent can be reused by multiple builds
Standby agent mode
Off
Standby agent mode off.
Manual
If you know your workloads' usage patterns, you can configure standby agents as per the scale conditions.
Automatic
If you don't know your usage patterns, you can rely on automatic forecasting based on past data.
Pre-provisioning quick starters
Start from scratch
Pre-provisioning TimeZone
(UTC) Coordinated Universal Time
Pre-provisioning scheme *
Start Time
End Time
Count
Monday 08:00
Friday 17:00
2
Max time to live *
7.00:00:00
Grace period *
0.00:30:00
Standby agent percentage
Ubuntu Server 24.04 LTS - x64 Gen 2
100%

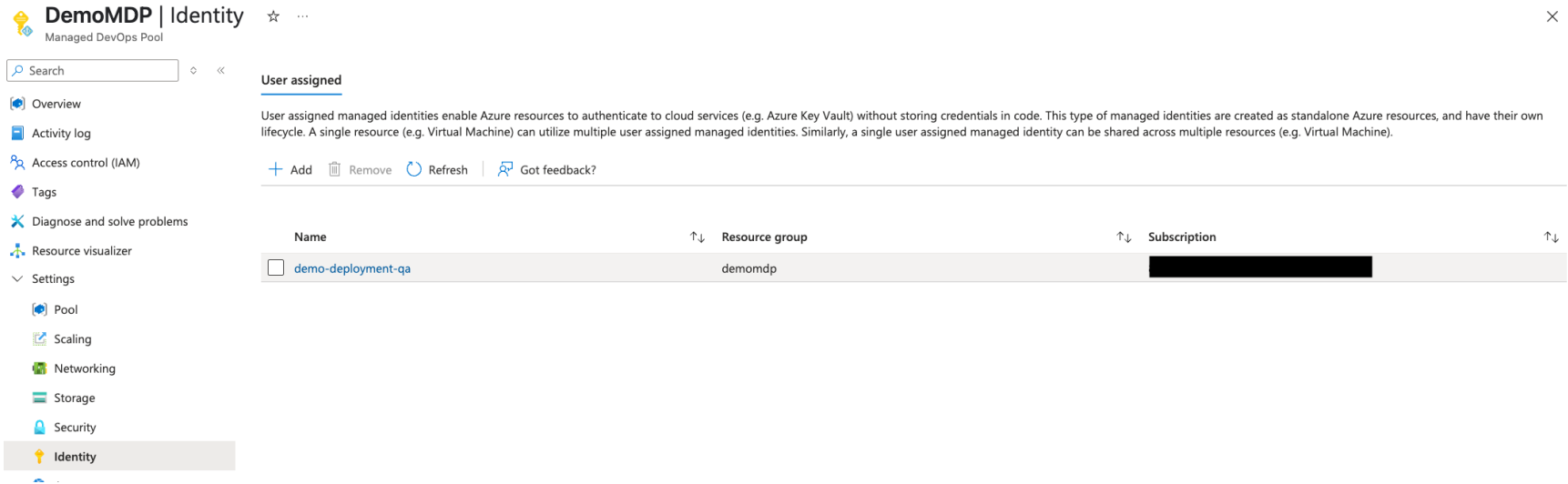
vNET integration



KeyVault Integration



Managed Identity



SECURITY

Threats

- Code injection
  - Hijacked NuGet / npm packages
  - Vulnerabilities / backdoors
- Broad network access
  - Agent can access all machines
  - Hole punching, remote access to target
- Local secrets
  - Secrets available to any pipeline
- No segregation
  - Dev/Test/Prod use the same infrastructure, same network

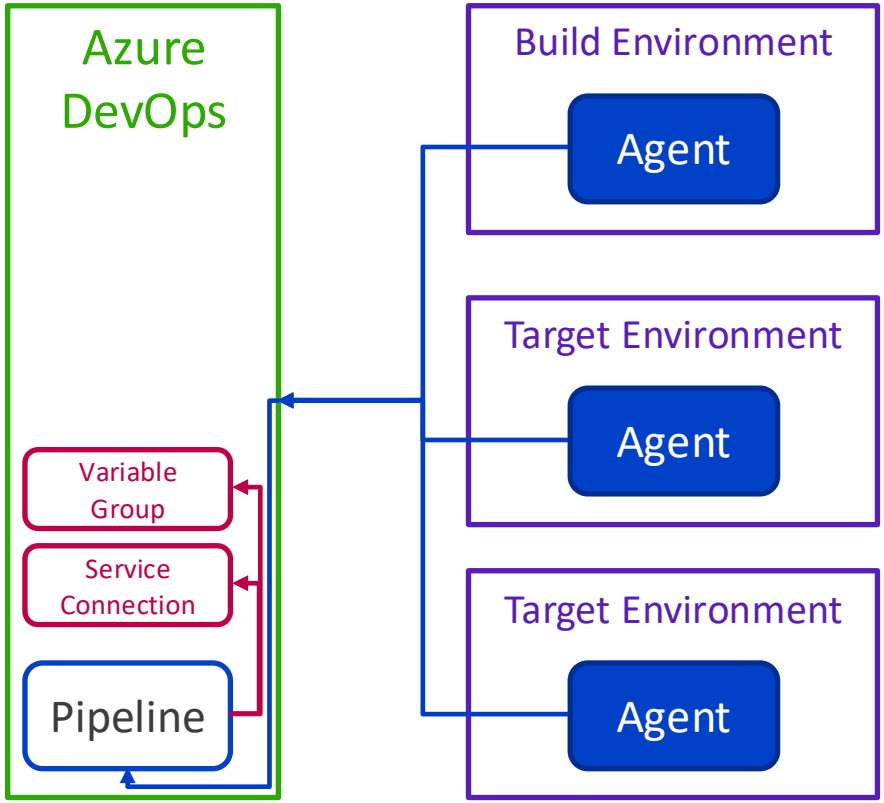
User Accounts / Permissions

- Agent Process
- Local user account running the agent process
- Access to local resources (files, network shares, other machines)

- Pipeline Context
- Communication with Azure DevOps and Access to its resources
- Security Principals
  - Project Collection Build Service
  - Project Build Service
- Integrates in Permission System of Azure DevOps

Agent Deployment

- Agent per Purpose (build, deploy target)
- Agent as near as possible to deployment target
- Connectivity from inside-out, no hole punching, isolate networking
- Least privileged
- Use secret from Azure DevOps, per environment vault instance
- Permit resource usage per pipeline

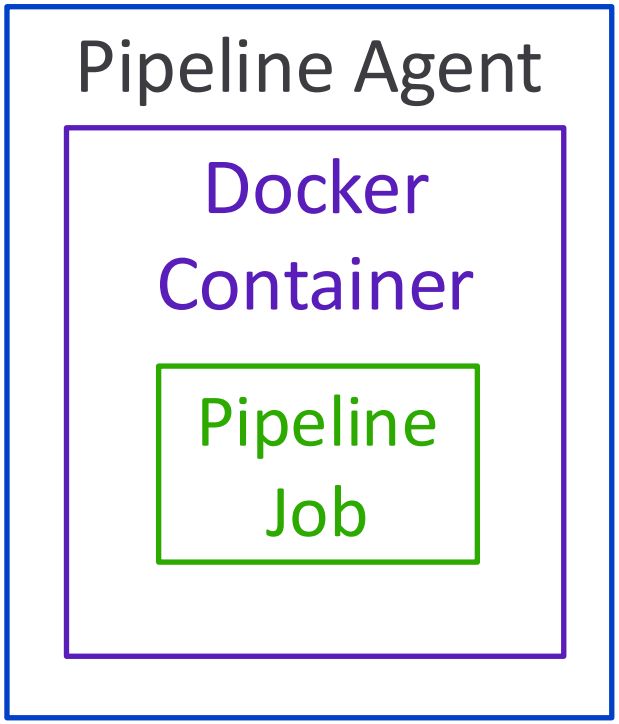


VM IMAGE VS. DOCKER CONTAINER

CI Tool Versioning Challenges

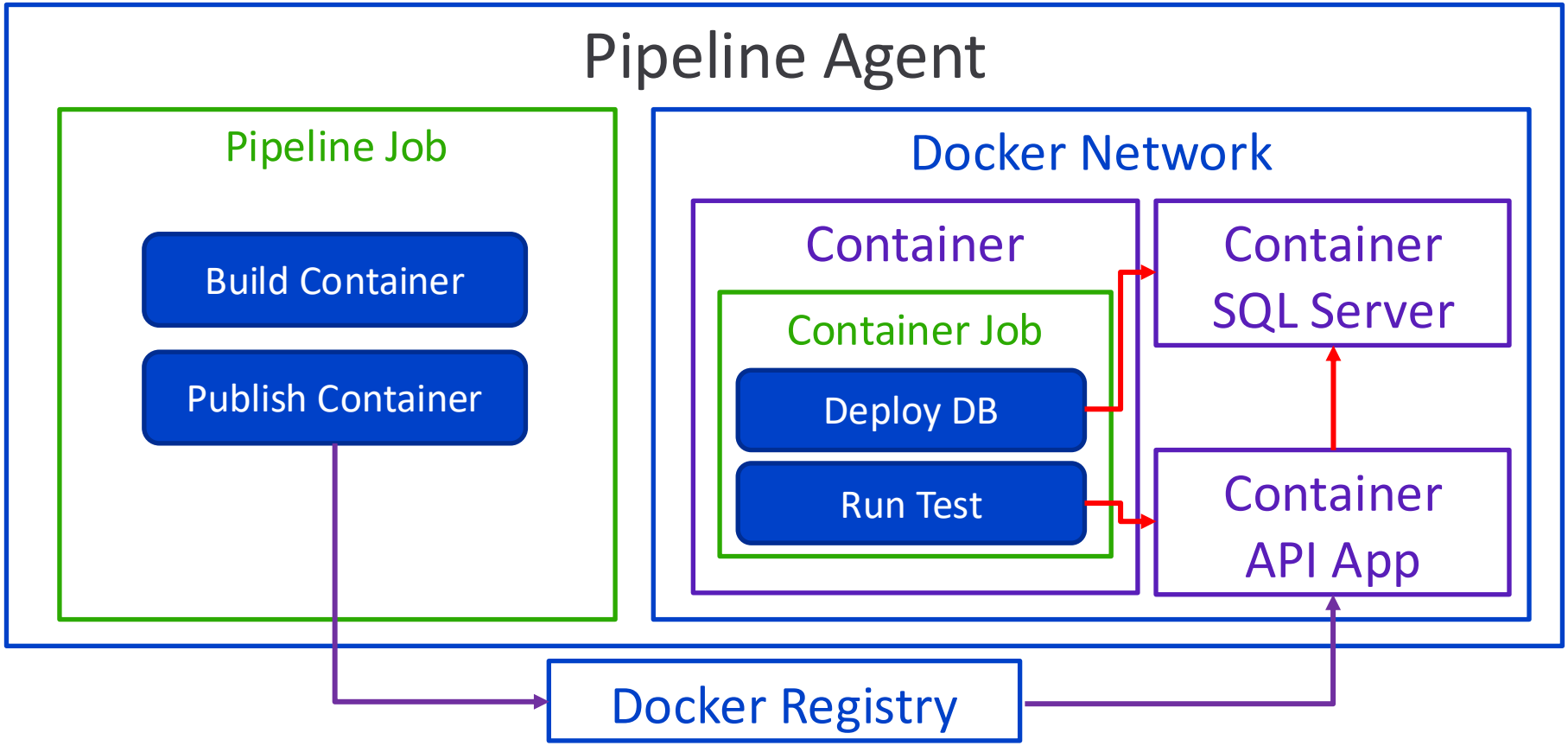
- Storing all VM images need a lot of storage
- Build VM snapshots not integrated in standard process
- Where to store the VM configuration?

Container Jobs



- Work with hosted pools and private pools
- Pipeline Job runs inside the container
- Each step runs inside the container

Bringing all together – Shift left



RECAP

Takeaways

- Managed DevOps Pool (MDP) – only advantages, no disadvantages (if Azure hosting fits)
- vNET integration satisfies all connectivity requirements
- Any VM image (Microsoft Images and custom images)
- Any hardware (which is supported by Azure)
- Managed Identity
- Flexible cloud scaling and billing

Thank you! Let’s continue the conversation

If you would like to discuss your challenges or explore how we can support you, please contact us.

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APPENDIX

Links

- https://www.developer-world.de/dwx-insights/cloud-devops/dw-managed-devops-pools
- https://learn.microsoft.com/en-us/azure/devops/managed-devops-pools/